2. Requirement Analysis Phase

2.1 Introduction

The Requirement Analysis Phase identifies, analyzes, and documents the requirements of LegalEase: AI-Powered Legal Document Generator. It focuses on understanding the problems faced by users in preparing legal documents and defining the features needed to solve them. LegalEase is designed as an intelligent drafting system that uses Google Gemini models to generate, explain, and review legal documents from simple user inputs.

2.2 Problem Analysis

Preparing a legal document is difficult for non-experts. The major problems identified are:

- High cost and limited availability of lawyers for routine documents.
- Generic templates that do not fit the user's specific situation.
- Legal language that is hard for non-experts to read and understand.
- Missing important clauses such as termination, payment terms, confidentiality, or dispute resolution.
- Errors in names, dates, and amounts when editing templates manually.
- Different requirements across jurisdictions.
- No easy way to save, revise, and export documents in a professional format.

2.3 Objectives of Requirement Analysis

- To identify the needs and expectations of users.
- To define the major functionalities of LegalEase.
- To identify the required hardware and software resources.
- To determine functional and non-functional requirements.
- To define requirements for Gemini-based integration.
- To provide a clear foundation for design and development.

2.4 Functional Requirements

2.4.1 User Registration and Authentication

- User registration and secure login.
- User profile management (name, email, default jurisdiction).
- Logout functionality.
- Basic authentication and input validation.

2.4.2 Document Selection and Input

- Provide a list of supported document types.
- Collect party details, dates, amounts, key terms, and jurisdiction through a guided form.
- Validate required fields before drafting.

2.4.3 AI Document Generation

- Generate a complete draft with title, parties, recitals, numbered clauses, and signature block using Gemini.
- Use the user-provided details exactly and mark any missing information as a placeholder.
- Allow the user to edit or regenerate the draft.

2.4.4 Clause Explanation

- Explain any selected clause in plain language.
- Highlight the obligations and rights of each party.

2.4.5 Review and Risk Analysis

- Check the draft for missing standard clauses, vague or ambiguous wording, and one-sided terms.
- Present suggestions that the user can accept or ignore.

2.4.6 Editing and Versions

- Edit the full document or an individual clause.
- Regenerate a single clause with additional instructions.
- Save versions of a document.

2.4.7 Export

- Preview the final document.
- Export as PDF or Word (.docx).

2.4.8 Document History

- Store saved documents and versions.
- Open, edit, duplicate, or delete previous documents.

2.5 Non-Functional Requirements

Requirement	Description
Usability	Simple, clean, responsive interface usable without legal or technical knowledge.
Performance	Draft generation should complete within a reasonable time, with loading indicators.
Security	Secure authentication, protected APIs, and the Gemini API key kept on the server.
Privacy	Documents contain personal information; users can access only their own documents, which are never shown to other users.
Content Safety	Requests for unlawful or harmful documents are blocked using validation and Gemini safety settings.
Accuracy	Drafts follow structured prompts and user-supplied facts; unknown details are left as placeholders.
Legal Disclaimer	The application states that output is a draft for information only and not legal advice.
Reliability	Accurate storage of data and graceful handling of API errors, timeouts, and blocked responses.
Maintainability	Modular design so new document types can be added easily.

2.6 Hardware Requirements

- Computer or laptop with minimum 4 GB RAM and 10 GB free storage.
- Internet connection (required for Gemini API access).
- Keyboard and mouse; smartphone or computer for accessing the application.

2.7 Software Requirements

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, FastAPI
Database	SQLite / PostgreSQL
AI / LLM	Google Gemini API
Authentication	JWT
Export	ReportLab and python-docx
IDE	Visual Studio Code
Version Control	Git and GitHub
Operating System	Windows / Linux / macOS

2.8 Expected Outcome

The phase produces a well-defined set of requirements that serves as the foundation for the planning, design, development, testing, and demonstration phases. LegalEase is expected to give users a convenient platform for creating, understanding, and exporting well-structured legal document drafts with AI support.